

ENDG 319



Probability, Statistics and Machine Learning

Lecture 7: Random Variables and Probability Distributions

PMF and CDF • Mean and variance • Continuous density

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Learning outcomes for Lecture 7

FROM OUTCOMES TO A NUMERICAL PROBABILITY MODEL

- 1 Define** Map an experimental outcome to a numerical value.
- 2 Describe** Construct and read a PMF, PDF, and CDF.
- 3 Summarize** Compute a mean, variance, and standard deviation.
- 4 Apply** Translate a specification into a probability or percentile.

Choose the variable first. Then sum probability masses or integrate density.

A random variable assigns a number to an outcome

THE SOURCE EXPERIMENT: CLASSIFY THREE ITEMS

Let X be the number of defective items. D = defective; N = nondefective.

Experimental outcome(s)	Value of X
NNN	0
DNN, NDN, NND	1
DDN, DND, NDD	2
DDD	3

Different outcomes can produce the same numerical value of X .

Outcome, random variable, value, and event

KEEP THESE FOUR IDEAS DISTINCT

Outcome

A result of the
experiment:
DNN

Random variable

The counting rule:
 X = number of defects

Observed value

Apply the rule to DNN:
 $x = 1$

$$\{X = 1\} = \{DNN, NDN, NND\}$$

“ $X = 1$ ” describes an event: all outcomes that the rule X maps to 1.

Example 14: sum the resistances in series

TARGET RESISTANCE: 30 Ω

Select one resistor from each labelled group. Let X be their total resistance.

Labelled 10 Ω

Actual values: 9, 10, 11 Ω

Labelled 20 Ω

Actual values: 19, 20, 21 Ω



$X = \text{first resistance} + \text{second resistance}$

Assume the nine pairs of individual resistors are equally likely.

Example 14: nine outcomes, but only five sums

EACH CELL IS ONE OUTCOME WITH PROBABILITY 1/9

First ↓ / Second →	19 Ω	20 Ω	21 Ω
9 Ω	28	29	30
10 Ω	29	30	31
11 Ω	30	31	32

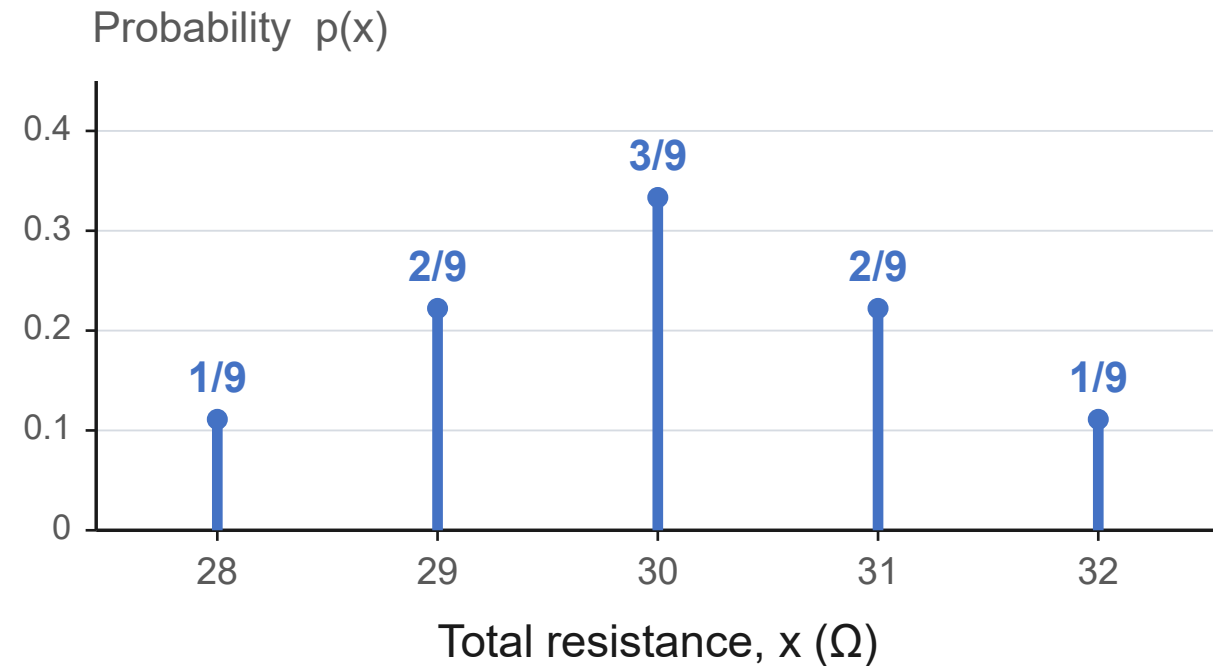
Each cell displays X, the total resistance in ohms.

The sums are not equally likely: 30 Ω appears in three cells; 28 Ω appears in one.

Example 14: group outcomes to obtain the PMF

ADD PROBABILITIES OF OUTCOMES WITH THE SAME SUM

$x (\Omega)$	$P(X = x)$
28	$1/9$
29	$2/9$
30	$3/9$
31	$2/9$
32	$1/9$



$$\{X = 29\} = \{(9, 20), (10, 19)\}$$

There are five possible values of X , but their probabilities are not all $1/5$.

Discrete and continuous random variables

USE THE MEASUREMENT OR COUNTING MODEL—NOT JUST A GROUPED TABLE

Discrete

Possible values can be listed.

Example: resistor totals
 $\{28, 29, 30, 31, 32\} \Omega$

Continuous

Possible values range over an interval.

Example: disk lifetime Z
measured in hours.

Grouping lifetimes into intervals does not make the underlying lifetime discrete.

Disk lifetime: intervals do not specify a full density

SOURCE EXAMPLE: Z IS THE NUMBER OF HOURS UNTIL FAILURE

Lifetime interval	Proportion
$0 \leq Z \leq 100 \text{ h}$	0.10
$100 < Z \leq 500 \text{ h}$	0.50
$Z > 500 \text{ h}$	0.40

Z is continuous under the stated lifetime model.

The table gives interval probabilities, not probabilities for individual exact lifetimes.

The probability mass function: one value at a time

FOR A DISCRETE RANDOM VARIABLE X

$$p(x) = P(X = x)$$

A PMF lists each possible value and the probability assigned to it.

Nonnegative masses

$p(x) \geq 0$ for every x

Total probability is one

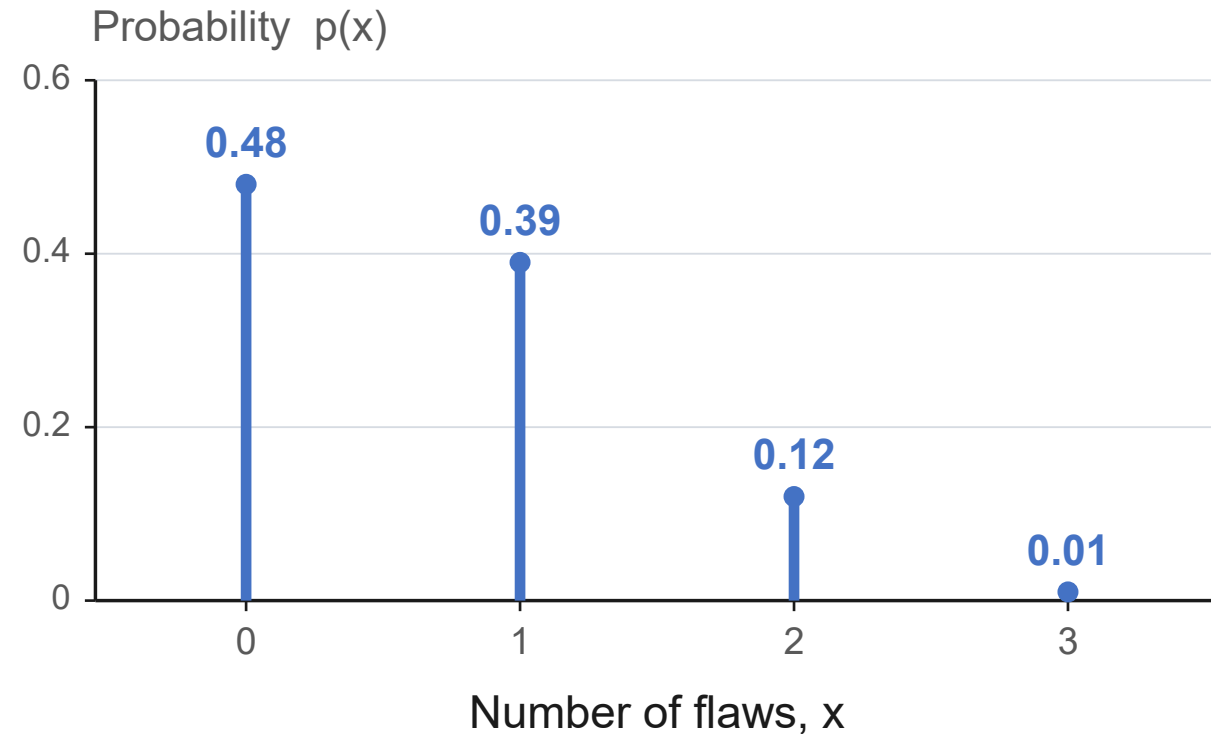
$$\sum_x p(x) = 1$$

A PMF can be presented as a table, a formula, or a graph.

Example 15: flaws in a one-inch copper wire

X = NUMBER OF FLAWS IN A RANDOMLY SELECTED PIECE

x	p(x)
0	0.48
1	0.39
2	0.12
3	0.01

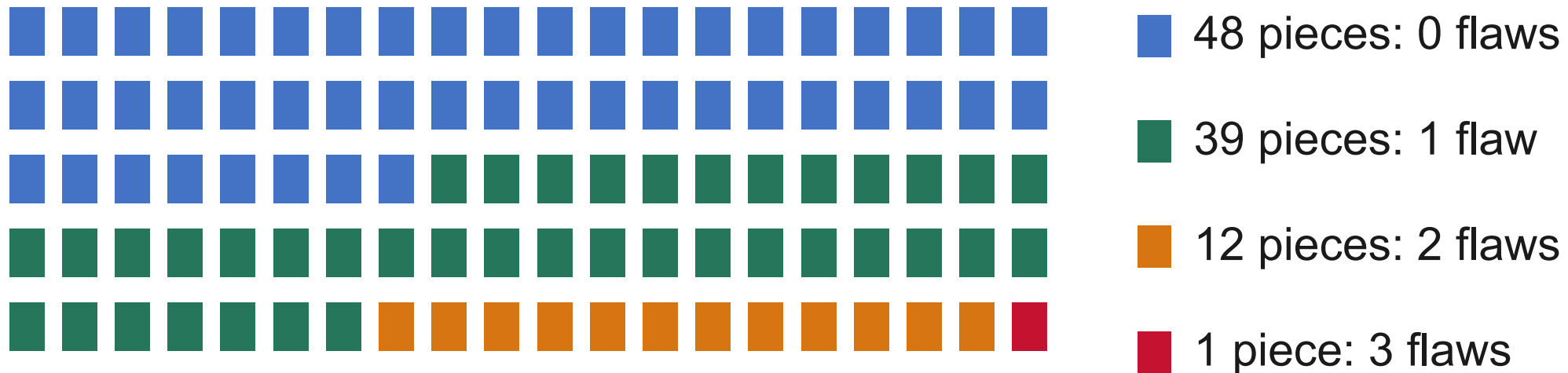


$$0.48 + 0.39 + 0.12 + 0.01 = 1$$

Here $p(0) = 0.48$: 48% of the population of wire pieces has no flaws.

A probability distribution describes a population

A HYPOTHETICAL 100-PIECE REPRESENTATION OF EXAMPLE 15



The population view links a probability model to the long-run proportions of numerical values.

This is a teaching representation of the PMF—not a newly collected sample.

Example 8 revisited: count burned-out bulbs

EIGHT GOOD BULBS, TWO BURNED-OUT BULBS; SELECT FOUR

Let X be the number of burned-out bulbs selected.



Burned-out selected: x	Good selected: $4 - x$
0	4
1	3
2	2

Select without replacement; every four-bulb subset of the ten individual bulbs is equally likely.

Example 8 revisited: the full PMF

COUNT FAVOURABLE FOUR-BULB SUBSETS FOR EACH x

$$p(x) = \frac{\binom{2}{x} \binom{8}{4-x}}{\binom{10}{4}}, \quad x = 0, 1, 2$$

x	Favourable subsets	$p(x)$
0	$1 \times 70 = 70$	$70/210 = 1/3$
1	$2 \times 56 = 112$	$112/210 = 8/15$
2	$1 \times 28 = 28$	$28/210 = 2/15$

$$\binom{10}{4} = 210, \quad 70 + 112 + 28 = 210$$

These three events partition all possible four-bulb selections.

PMF versus CDF: exact value or accumulated probability?

THE CDF IS DEFINED FOR EVERY REAL THRESHOLD x

Probability mass function

Question: “Exactly x ?”

$$p(x) = P(X = x)$$

One mass at one possible value.

Cumulative distribution (CDF)

Question: “At most x ?”

$$F(x) = P(X \leq x)$$

All masses at values $\leq x$.

“At most” includes equality: the CDF includes the mass at its threshold.

Evaluate a discrete CDF by collecting possible values

EXAMPLE 15: $P(0)=0.48$, $P(1)=0.39$, $P(2)=0.12$, $P(3)=0.01$

$$F(x) = \sum_{t \leq x} p(t)$$

Threshold x	Values t included	F(x)
1	0, 1	$0.48 + 0.39 = 0.87$
1.5	0, 1	0.87
2.6	0, 1, 2	$0.48 + 0.39 + 0.12 = 0.99$

The threshold can be non-integer; X still takes only its allowed integer values.

Build the complete CDF one interval at a time

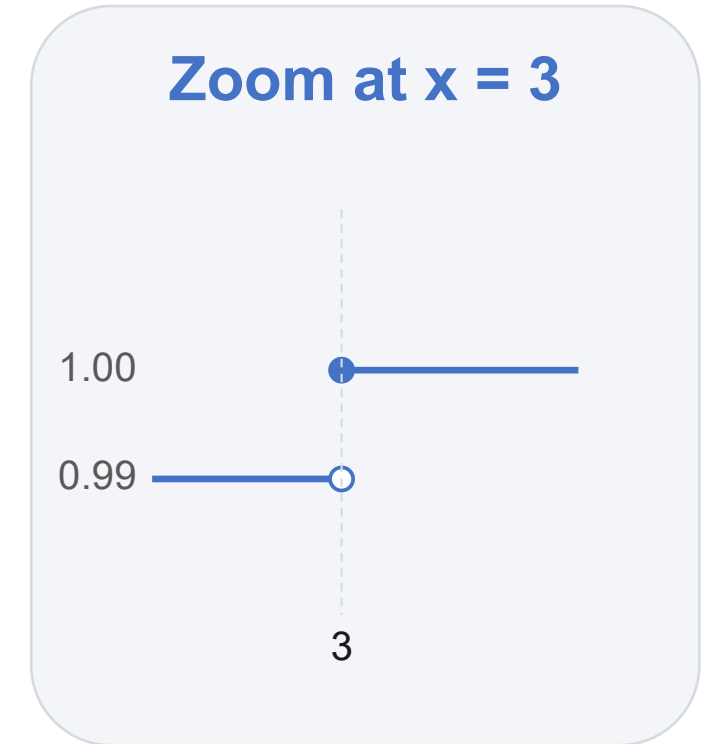
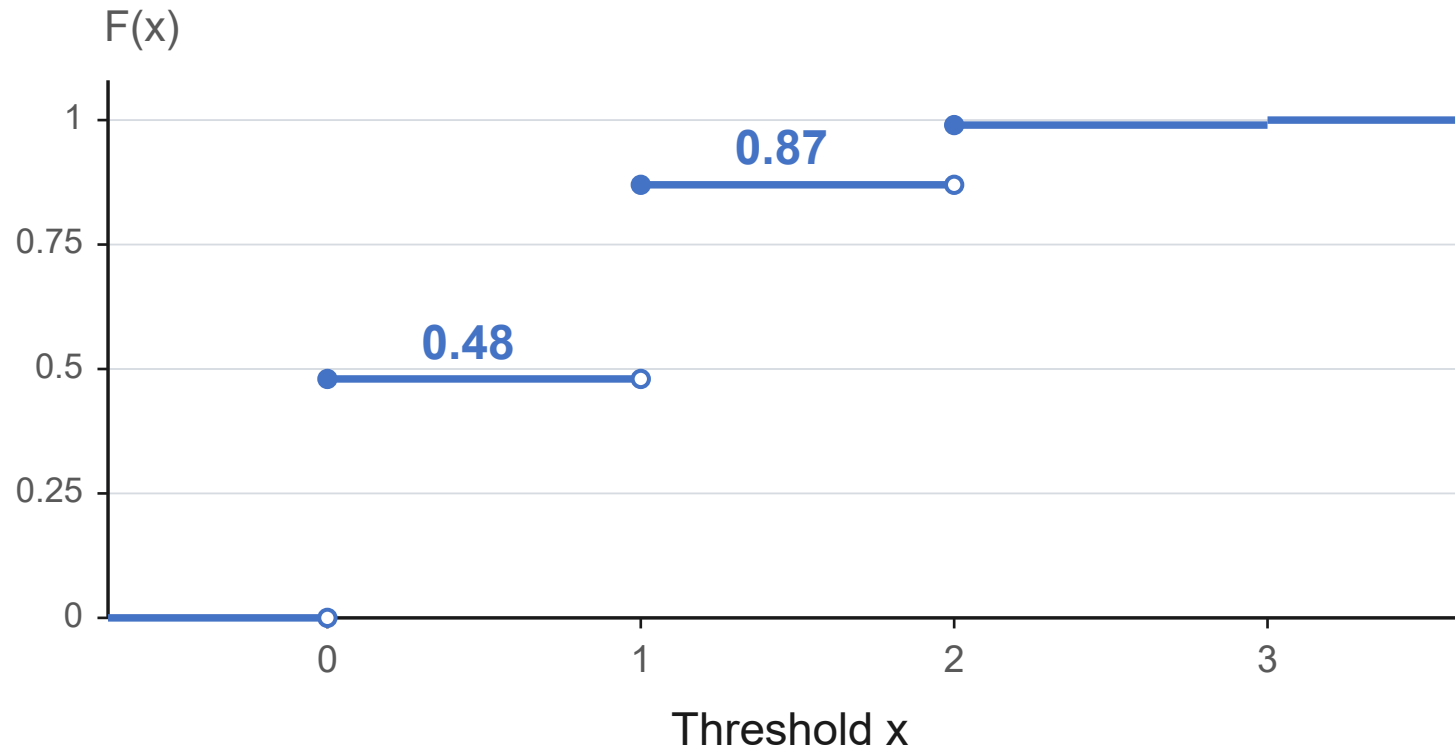
EXAMPLE 15: ACCUMULATE THE WIRE-FLAW PROBABILITIES

Threshold interval	Values included	F(x)
$x < 0$	None	0
$0 \leq x < 1$	0	0.48
$1 \leq x < 2$	0, 1	0.87
$2 \leq x < 3$	0, 1, 2	0.99
$x \geq 3$	0, 1, 2, 3	1

For $x < 0$, $\{X \leq x\}$ is the empty event, so $F(x) = 0$.

A discrete CDF is a step function

FILLED ENDPOINT: VALUE INCLUDED | OPEN ENDPOINT: VALUE EXCLUDED



At $x = 3$, $F(3) = 1$. The final jump is small because $p(3) = 0.01$.

CDF checkpoint: translate the inequality first

USE EXAMPLE 15; DISCUSS BEFORE LOOKING AT THE NEXT SLIDE

- A** Find $p(1.5)$ and $F(1.5)$.
- B** Find $P(X < 1)$ and $P(X \leq 1)$.
- C** Find $P(X > 1)$.
- D** Find $P(0 < X < 3)$.

Start by listing which of 0, 1, 2, and 3 satisfy the event.

CDF checkpoint: solutions and interpretation

ONE PMF, FOUR DIFFERENT EVENT DESCRIPTIONS

Part	Answer	Reason
A	$p(1.5) = 0; F(1.5) = 0.87$	1.5 is impossible; 0 and 1 are ≤ 1.5 .
B	0.48; 0.87	Including 1 adds its mass of 0.39.
C	$1 - 0.87 = 0.13$	$X > 1$ is the complement of $X \leq 1$.
D	$0.39 + 0.12 = 0.51$	Only $X = 1$ and $X = 2$ qualify.

For a discrete variable, changing $<$ to $\leq$ can change the probability.

The mean is a probability-weighted average

RETURN TO THE HYPOTHETICAL 100-PIECE POPULATION

Number of flaws	Number of pieces	Total flaws
0	48	0
1	39	39
2	12	24
3	1	3

$$\mu = \frac{0(48) + 1(39) + 2(12) + 3(1)}{100} = 0.66$$

An average of 0.66 flaws does not mean that any one piece has 0.66 flaws.

Expected value of a discrete random variable

MEAN, EXPECTATION, AND EXPECTED VALUE REFER TO THE SAME QUANTITY

$$\mu_X = E(X) = \sum_x x p(x)$$

Multiply each possible value by its probability, then add.

$$E(X) = 0(0.48) + 1(0.39) + 2(0.12) + 3(0.01) = 0.66$$

The probabilities supply the weights; do not average the possible values equally.

Variance and standard deviation of a discrete variable

PROBABILITY-WEIGHT THE SQUARED DISTANCES FROM THE MEAN

$$\sigma_X^2 = V(X) = \sum_x (x - \mu_X)^2 p(x)$$

Equivalent computational form

$$\sigma_X^2 = \sum_x x^2 p(x) - \mu_X^2$$

$$\sigma_X = \sqrt{\sigma_X^2}$$

Use probability weights—not the sample-variance denominator $n - 1$.

Example 16: weekly process recalibration

X = NUMBER OF RECALIBRATIONS IN ONE WEEK

The process is recalibrated whenever output quality falls below specifications.

x	0	1	2	3	4
p(x)	0.35	0.25	0.20	0.15	0.05

Find the mean and standard deviation of X.

Work in this order: $E(X)$, then $E(X^2)$, then variance, then standard deviation.

Example 16: a calculation table prevents omissions

KEEP FULL PRECISION UNTIL THE FINAL SQUARE ROOT

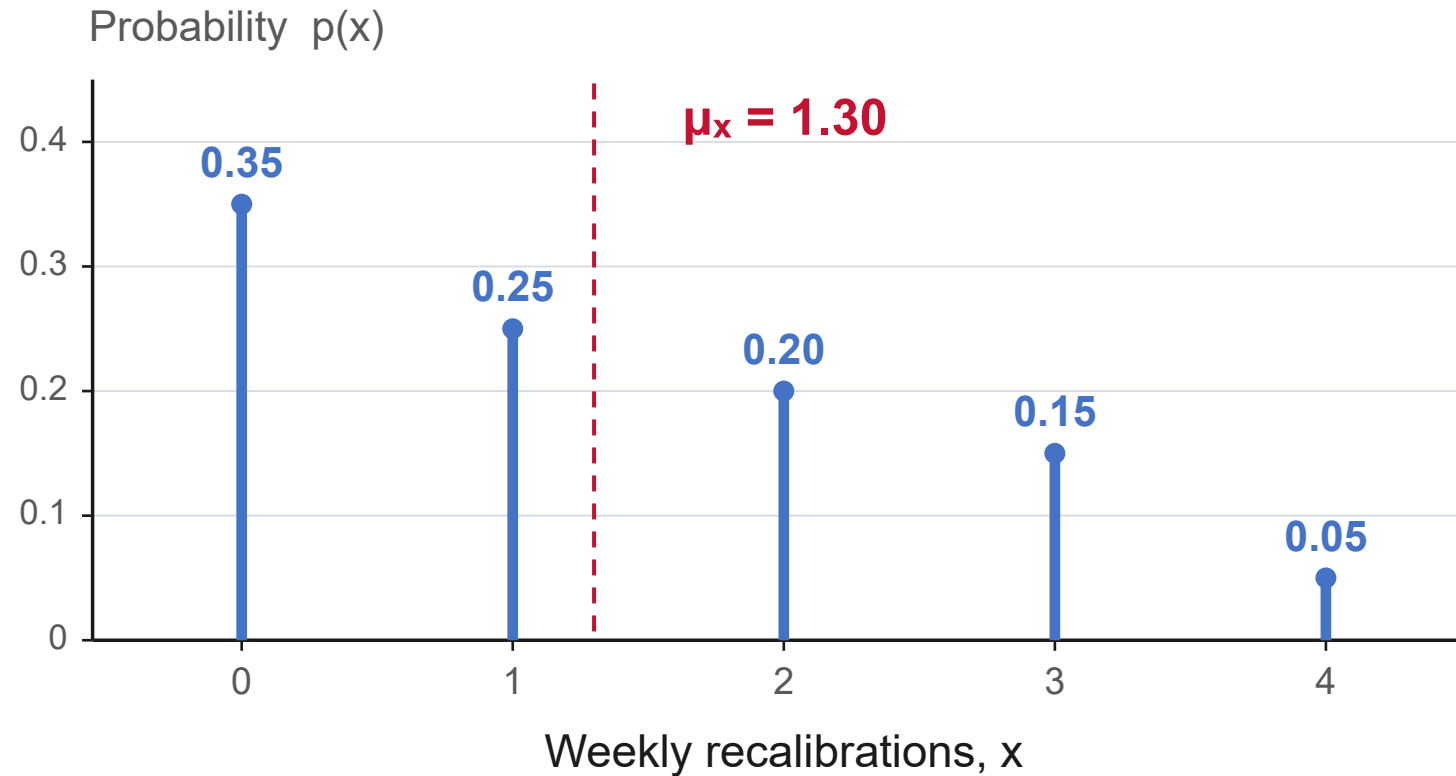
x	p(x)	x p(x)	x ² p(x)
0	0.35	0.00	0.00
1	0.25	0.25	0.25
2	0.20	0.40	0.80
3	0.15	0.45	1.35
4	0.05	0.20	0.80
Sum	1.00	1.30	3.20

$$\mu_X = 1.30, \quad \sigma_X^2 = 3.20 - (1.30)^2 = 1.51$$

$\sigma_X = \sqrt{1.51} \approx 1.23$ recalibrations per week.

Example 16: the mean as a balance point

PROBABILITY MASSES DETERMINE THE BALANCE—NOT EQUAL WEIGHTS



Model summaries

Mean: 1.30

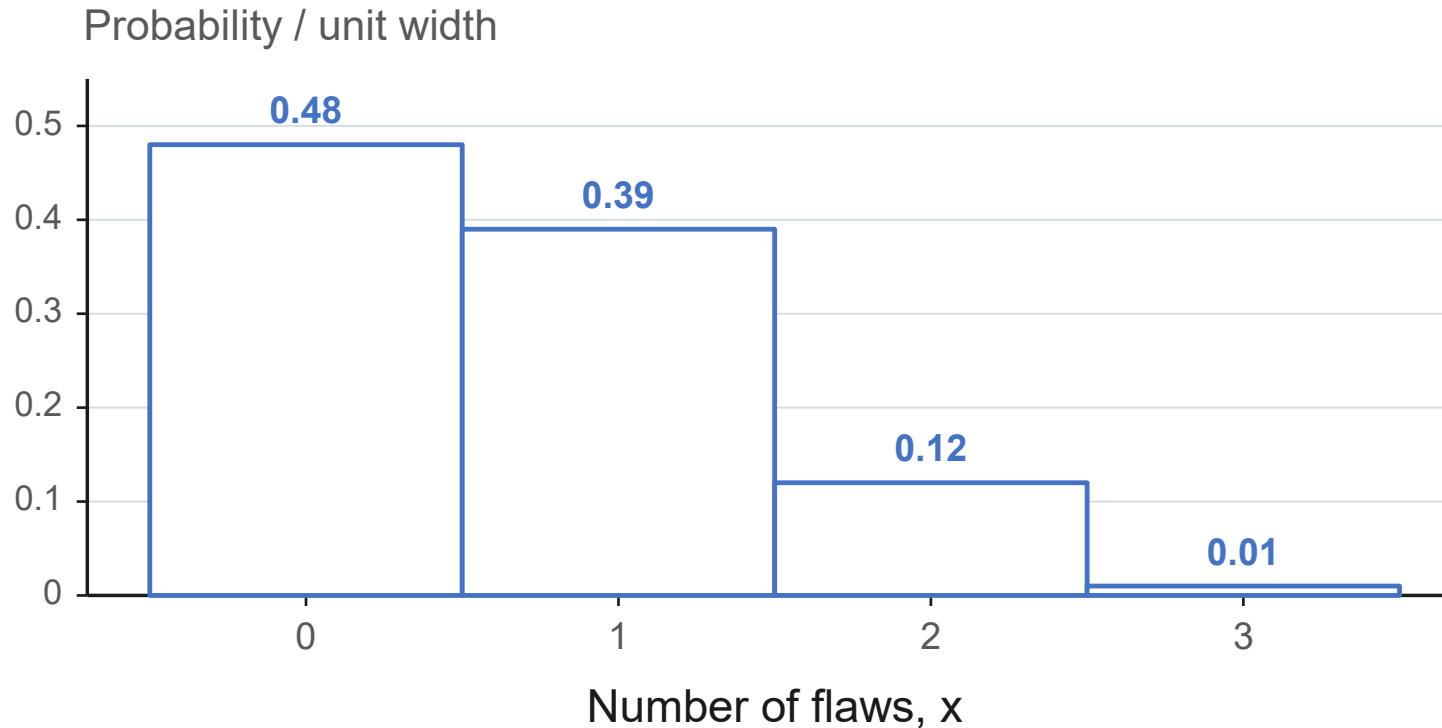
Variance: 1.51

SD: about 1.23

The most likely count is 0; the mean is 1.30. They answer different questions.

A probability histogram makes mass into area

FOR EVENLY SPACED VALUES: CENTRE ONE RECTANGLE AT EACH VALUE



Construction

Width = spacing = 1

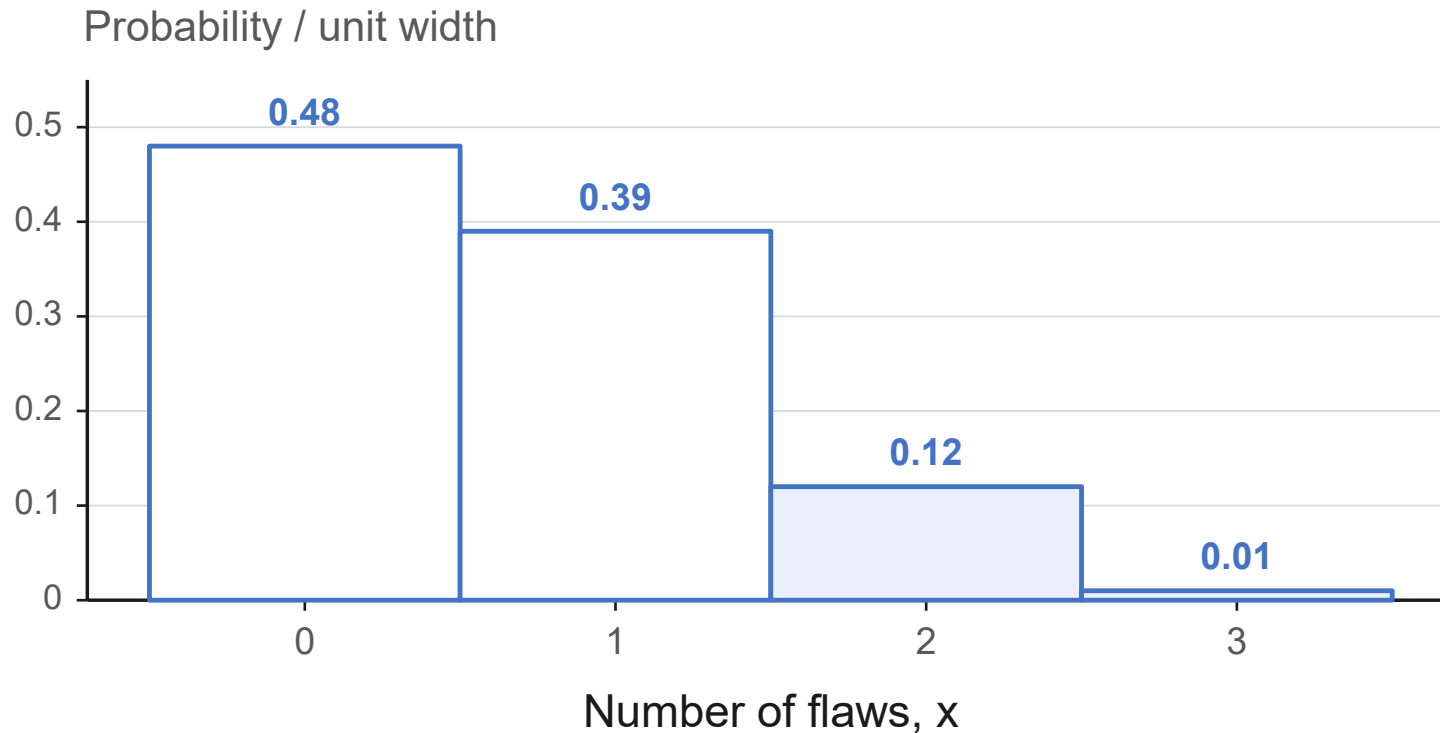
Area = $p(x)$

Height = $p(x) / \text{width}$

Unit width makes the numerical bar height equal to its probability.

Read the shaded area as an event probability

EXAMPLE 15: $X > 1$ INCLUDES EXACTLY THE VALUES 2 AND 3



$$P(X > 1) = p(2) + p(3)$$

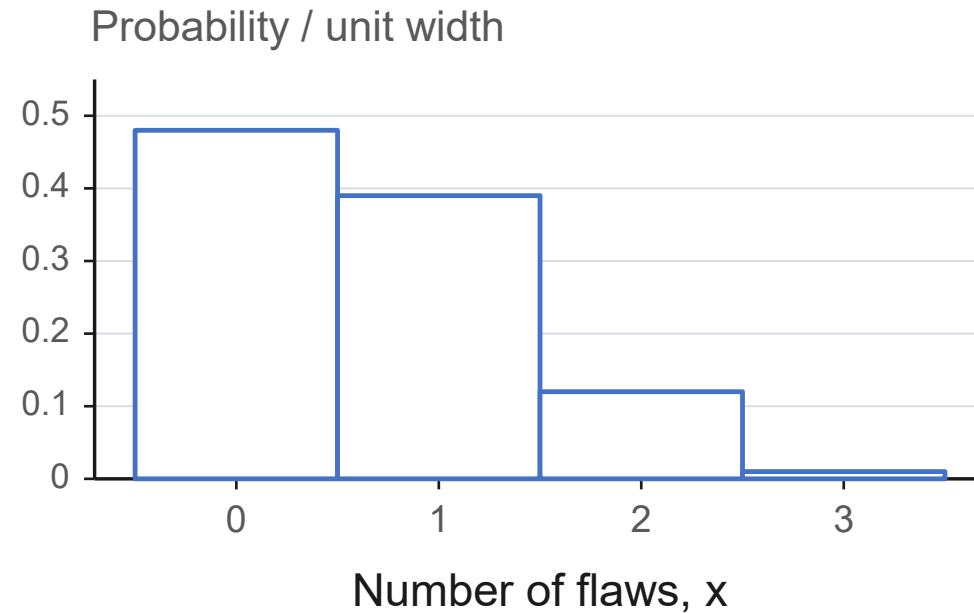
$$= 0.12 + 0.01 = 0.13$$

Shade the whole rectangles centred at the values in the event.

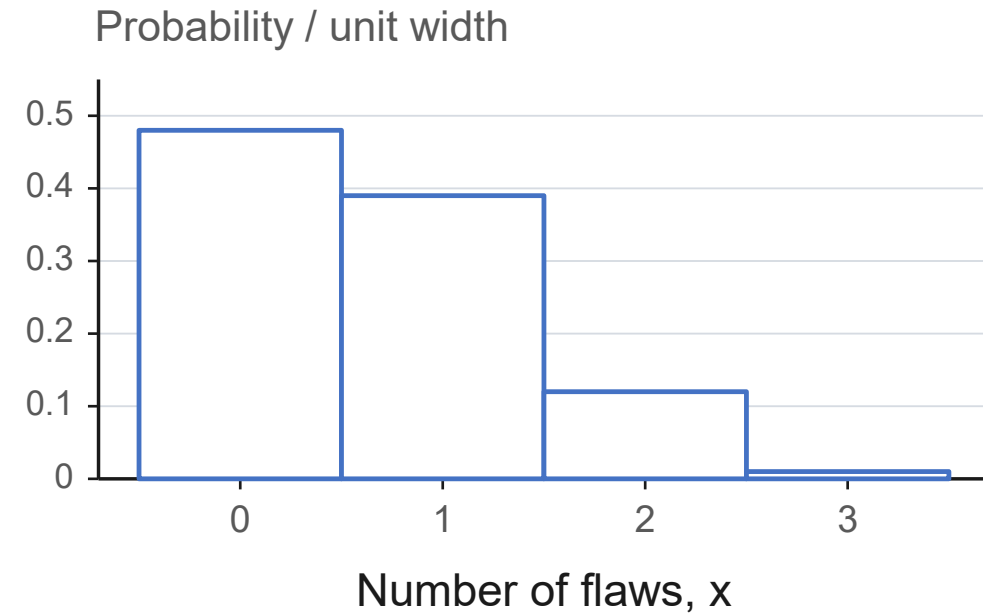
Histogram practice: identify the values, then shade

TRY BOTH SOURCE QUESTIONS BEFORE REVEALING THE ANSWERS

A. $P(X = 1)$



B. $P(0 < X < 3)$

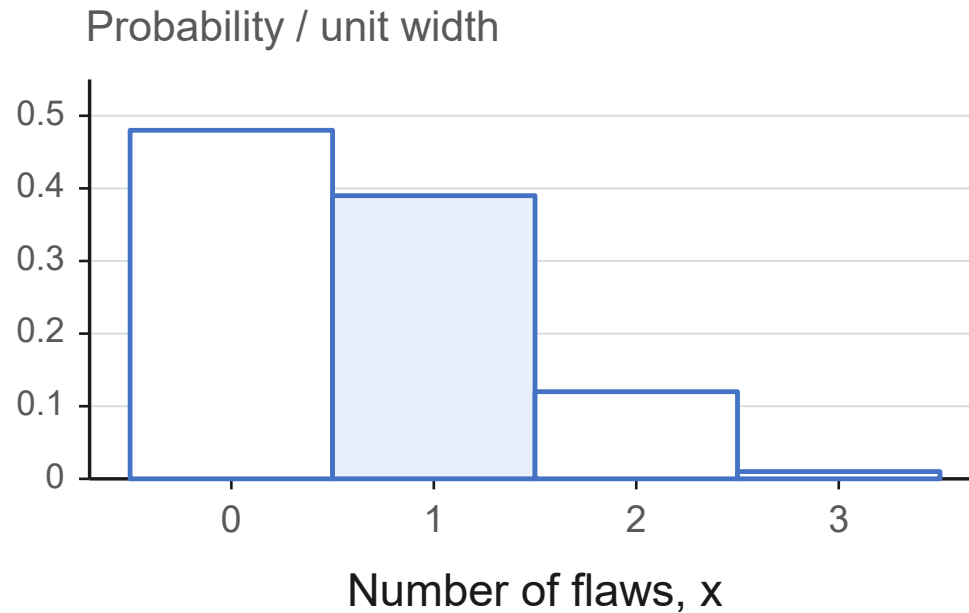


First identify the values in the event; then shade their complete probability bars.

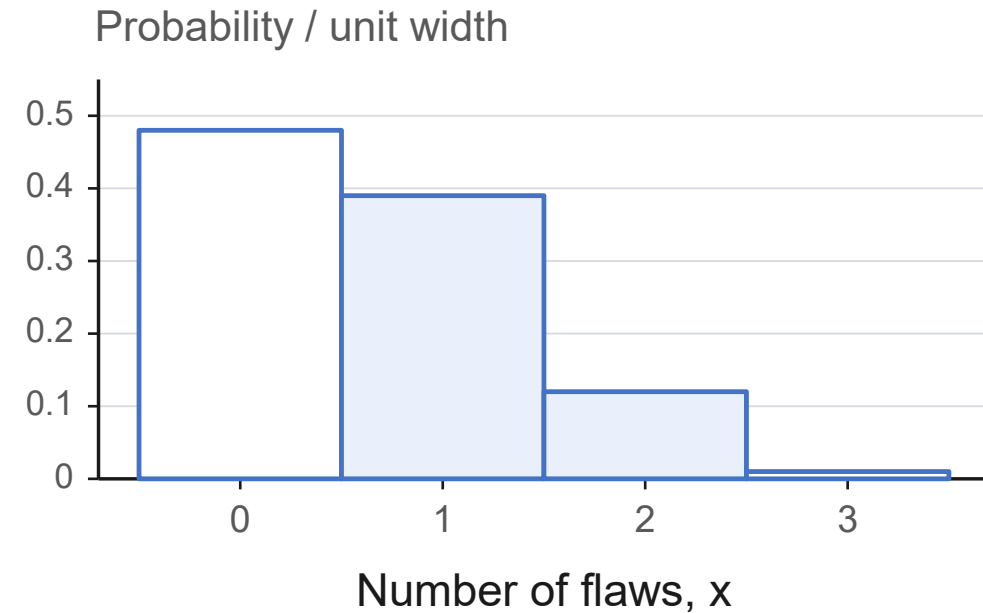
Histogram practice: complete solutions

STRICT ENDPOINTS EXCLUDE THE BARS AT 0 AND 3 IN PART B

A. $P(X = 1) = 0.39$



B. $P(0 < X < 3) = 0.51$



Part A uses $p(1)$. Part B uses $p(1) + p(2) = 0.39 + 0.12$.

Continuous variables: a range rather than a list

USE THE POSSIBLE-VALUE RANGES GIVEN IN THE SOURCE

$X = \text{water-bath temperature (}^{\circ}\text{C): } 39 \leq X \leq 41$



$Y = \text{assembly time (s): } 50 \leq Y < 70$



A range of possible values does not, by itself, specify a probability density.

Continuous events are sets of possible values

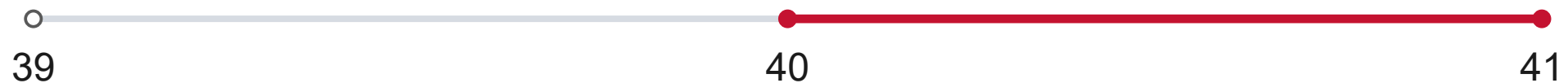
INTERVALS ARE IMPORTANT EXAMPLES OF EVENTS

Source range: $39 \leq X \leq 41$ °C

Event A: $39.5 \leq X \leq 40.5$



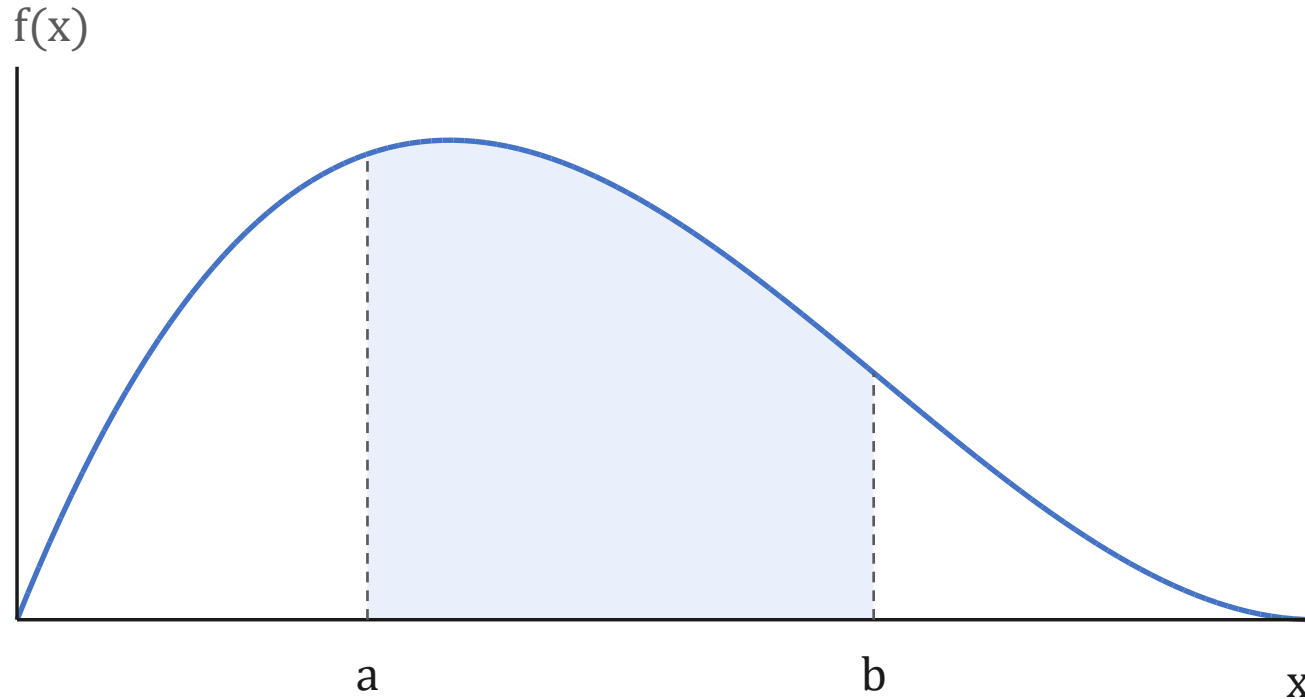
Event B: $40 \leq X \leq 41$



Define the event first. Its probability requires a model for how values are distributed.

A probability density function turns intervals into areas

THE PROBABILITY IS THE AREA—NOT THE CURVE HEIGHT



Schematic density curve; no numerical distribution is being assigned.

$$P(a \leq X \leq b)$$

$$= \int_a^b f(x) dx$$

Shaded region:
values between a and b

For a continuous variable with a PDF, add density over an interval by integrating.

Two conditions that a PDF must satisfy

A VALID DENSITY IS NONNEGATIVE AND HAS TOTAL AREA ONE

1. Nonnegative density

Density may vary with x , but it cannot be negative.

$$f(x) \geq 0$$

2. Total probability = 1

The area over the entire real line is one.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$f(x)$ is a density value. $P(a \leq X \leq b)$ is a probability.

The bound “at most 1” applies to probabilities, not necessarily to PDF heights.

Why including an endpoint does not change the area

FOR THE CONTINUOUS DISTRIBUTIONS REPRESENTED BY A PDF

$$P(X = a) = 0$$

A single point has zero width, so it contributes zero area.

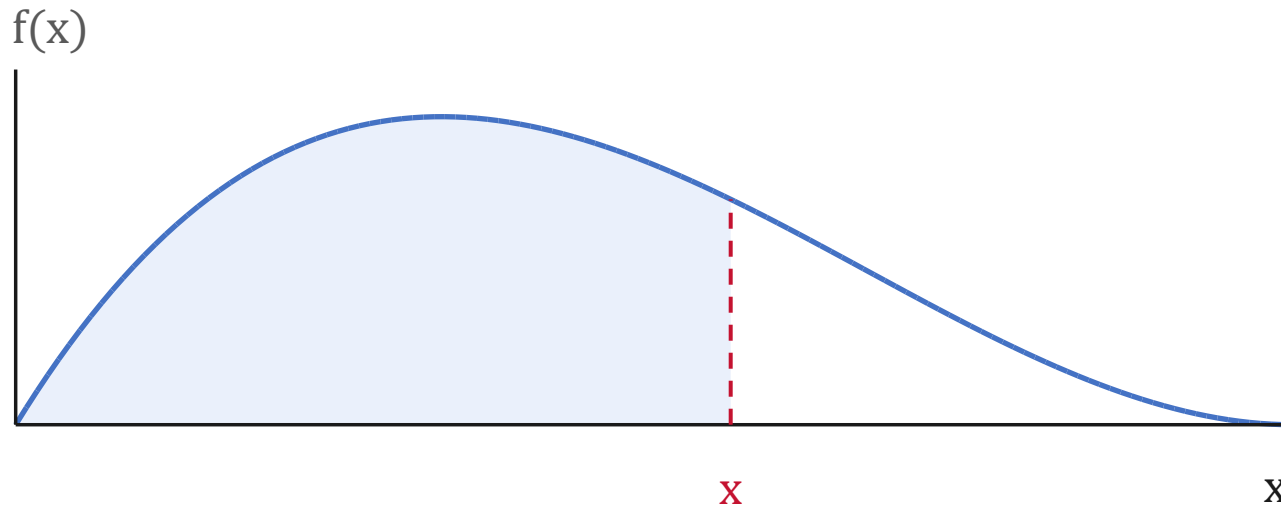
$$\begin{aligned} P(a \leq X \leq b) &= P(a < X \leq b) = P(a \leq X < b) = P(a < X < b) \\ &= \int_a^b f(x) dx \end{aligned}$$

Do not transfer this rule to a discrete PMF: there, a point can have positive mass.

The continuous CDF accumulates area from the left

THE DEFINITION IS STILL $F(x) = P(X \leq x)$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$



Interval probability

$$\begin{aligned} P(a < X \leq b) \\ &= F(b) - F(a) \end{aligned}$$

Use t inside the integral: x is the upper limit, not the variable being integrated.

Mean and variance: replace sums with integrals

THE SAME PROBABILITY-WEIGHTED LOGIC APPLIES

Mean / expected value

$$\mu_X = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Variance

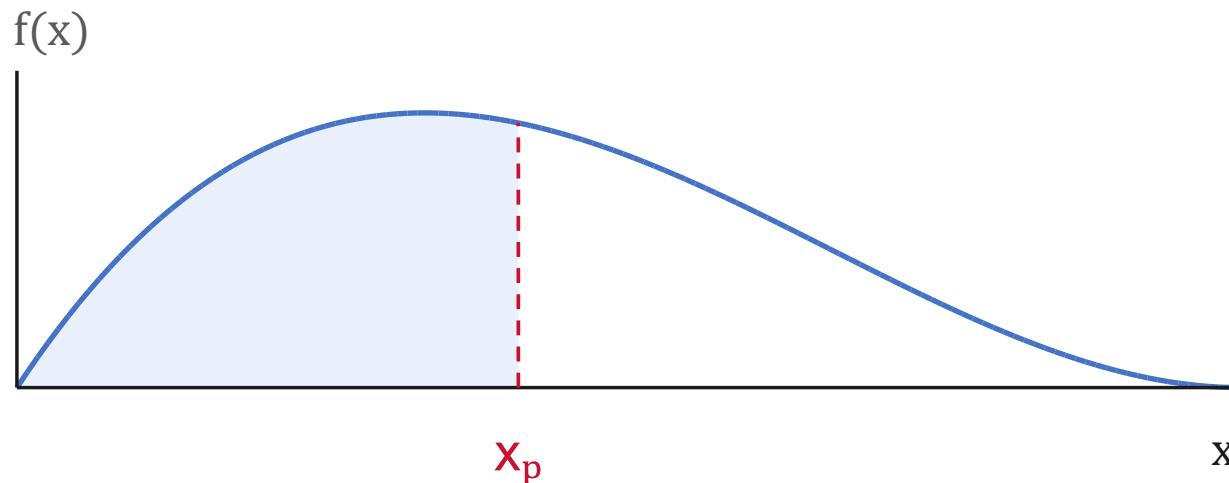
$$\begin{aligned}\sigma_X^2 &= \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x) dx \\ &= \int_{-\infty}^{\infty} x^2 f(x) dx - \mu_X^2\end{aligned}$$

Standard deviation is still $\sigma_x = \sqrt{(\sigma_x^2)}$. Use the density's actual support for calculation.

Percentiles locate a specified cumulative probability

THE PTH PERCENTILE LEAVES P% OF THE PROBABILITY TO ITS LEFT

$$F(x_p) = P(X \leq x_p) = \int_{-\infty}^{x_p} f(x) dx = \frac{p}{100}$$



Schematic: shaded area = $p/100$

Median: 50th percentile

Find x_m such that $F(x_m) = 0.50$.

A percentile is a value of X . The fraction $p/100$ is a probability.

Example 17: clearance and a scrap specification

X = HOLE RADIUS – SHAFT RADIUS, MEASURED IN MILLIMETRES

A shaft is inserted into a drilled hole. The clearance has the following density:

$$f(x) = 1.25(1 - x^4) \quad \text{for } 0 < x < 1$$

$$f(x) = 0 \text{ otherwise}$$

1. Scrap probability

Find $P(X > 0.8)$.

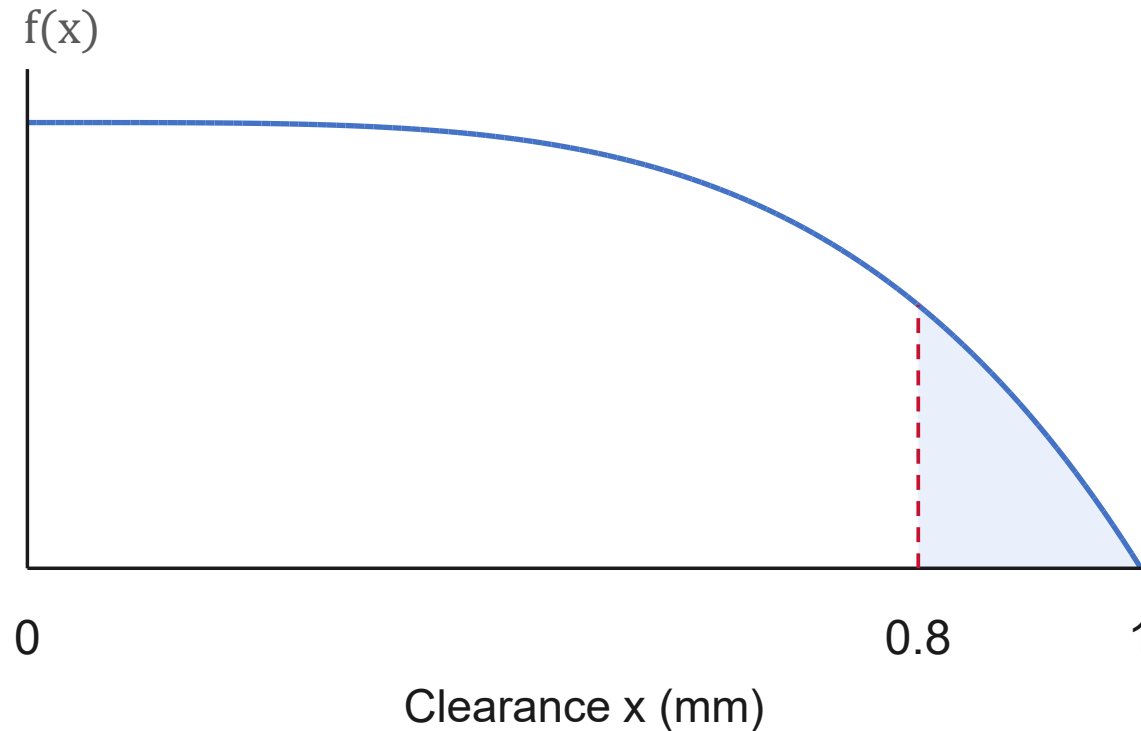
2. Cumulative probability

Find $F(x)$ for every real x .

For this exercise, any clearance greater than 0.8 mm requires scrapping the component.

Example 17: integrate the scrap tail

WORKED SOLUTION DERIVED FROM THE SUPPLIED DENSITY



$$\begin{aligned} P(X > 0.8) &= \int_{0.8}^1 1.25(1 - x^4) dx \\ &= [1.25x - 0.25x^5]_{0.8}^1 \\ &= 0.08192 = 8.192\% \end{aligned}$$

The model predicts that about 8.19% of components exceed the clearance limit.

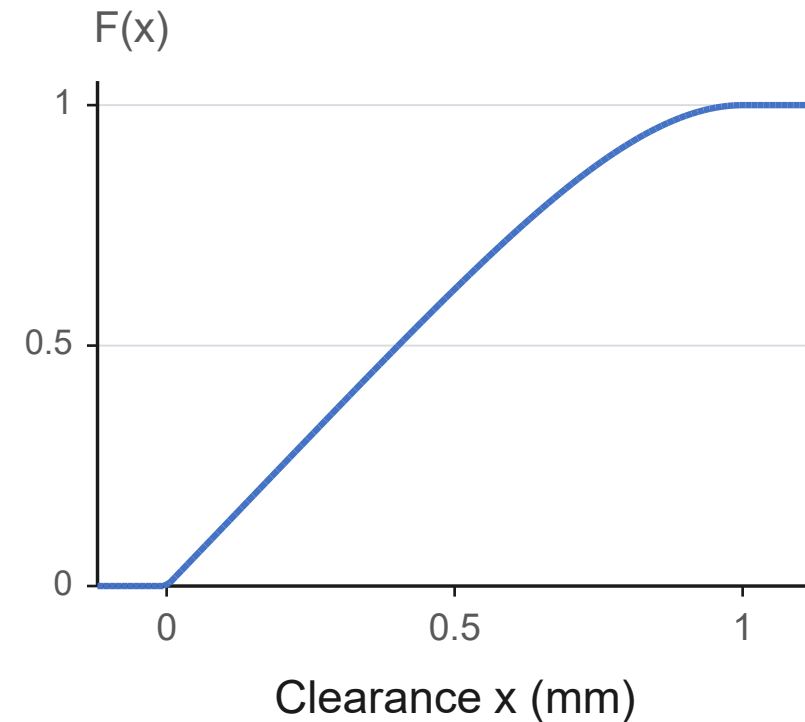
Example 17: give the CDF on the entire real line

WORKED SOLUTION DERIVED FROM $F(x) = P(X \leq x)$

Range of x	$F(x)$
$x \leq 0$	0
$0 < x < 1$	$1.25x - 0.25x^5$
$x \geq 1$	1

For $0 < x < 1$:

$$F(x) = \int_0^x 1.25(1 - t^4) dt$$



Check: $F(0.8) = 0.91808$, so $1 - F(0.8) = 0.08192$.

PMF, PDF, and CDF: a comparison to keep

USE THE SYMBOL AND THE EVENT STATEMENT TOGETHER

Question	Discrete model	Continuous model with PDF
Exactly x ?	$p(x) = P(X = x)$	$P(X = x) = 0$
At most x ?	$F(x) = \text{sum of masses at } t \leq x$	$F(x) = \text{area under } f \text{ to the left of } x$
What does the plot show?	PMF: probability at each value	PDF: density; interval area is probability
How do we find a mean?	Probability-weighted sum	Density-weighted integral

The CDF always reports a probability. A PDF height does not report an exact-value probability.

Final concept check

EXPLAIN EACH ANSWER IN WORDS BEFORE CALCULATING

- 1 For Example 15, is $F(2.6)$ a probability at $X = 2.6$?
- 2 Can a discrete random variable have a non-integer mean?
- 3 For a continuous X with a PDF, is $f(0.8)$ the probability that $X = 0.8$?
- 4 A clearance is scrapped above 0.8 mm. Which CDF expression gives the scrap proportion?

Use the wire model and clearance model already developed—no new distribution is needed.

Lecture 7: key takeaways and checkpoint answers

A RANDOM VARIABLE TURNS EXPERIMENTAL OUTCOMES INTO A NUMERICAL MODEL

- | | |
|--------------------------------|--|
| Random variable | Different experimental outcomes can map to the same value. |
| Discrete distribution | Add PMF masses; a CDF accumulates all masses at or below x . |
| Mean and variability | Use probability weights; SD is the square root of variance. |
| Continuous distribution | Integrate density; point probabilities are zero under a PDF. |

Checkpoint: 1. No; $F(2.6) = 0.99$. 2. Yes. 3. No. 4. $1 - F(0.8)$.

Questions? Start with the event in words, then choose $p(x)$, $f(x)$, or $F(x)$.